

Foundation (Jalapeno)

- Q1.** What is the slope of the line $y = 2x + 1$?
(A) 0
(B) 1
(C) 2
(D) -1
(E) -2
- Q2.** A spaceship travels 300 miles in 5 hours. What is the rate of change of distance with respect to time?
(A) 50 miles/hour
(B) 60 miles/hour
(C) 70 miles/hour
(D) 80 miles/hour
(E) 90 miles/hour
- Q3.** The graph of $y = mx + b$ has a slope of -3 . What is the rate of change of y with respect to x ?
(A) 3
(B) -3
(C) 0
(D) 1
(E) -1
- Q4.** A lab experiment measures the temperature of a substance over time, resulting in the equation $T = 2t - 5$. What is the slope of this line?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q5.** The table below shows the position of a planet over time.
- Q6.** The equation $y = -x + 2$ represents a line with a slope of
(A) 1
(B) -1
(C) 0
(D) 2
(E) -2
- Q7.** What is the slope of a line that is perpendicular to the line $y = 3x - 2$?
(A) $\frac{1}{3}$
(B) $-\frac{1}{3}$
(C) 0
(D) 3
(E) -3
- Q8.** A line passes through the points $(2, 3)$ and $(4, 5)$. What is the slope of this line?
(A) 1
(B) -1
(C) 0
(D) 2

Open Ended Questions (FRQ)

- Q9.** The cost of launching a satellite into space is directly proportional to the weight of the satellite. If it costs 2000 to launch a 2-ton satellite, what is the slope of the line that represents the cost of launching a satellite as a function of its weight?
- Q10.** A physicist is studying the motion of a particle along a straight line. The equation of motion is given by $s = 2t^2 + 3t - 4$, where s is the position of the particle and t is time. Find the slope of the line that best fits the data if the physicist takes measurements at $t = 0$, $t = 1$, and $t = 2$.

Chef's Tips

- Tip 1: Allow students to use calculators to check their answers
- Tip 2: Encourage students to use graphical representations to visualize the problems
- Tip 3: Remind students to label their axes and units when graphing